THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The HCF of 24, 36 and 60 is: 1

(a) 12 **(b)** 14 **(c)** 11 **(d)** 13

2. The LCM of 6, 8 and 12 is: 1

(a) 25 **(b)** 57 **(c)** 24 **(d)** 23

3. The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is: 1

(a) 13 **(b)** 14 **(c)** 15 **(d)** 12

4. The least number which when divided by 6, 8 and 12 leaves remainder 3 each time is: 1

(a) 24 **(b)** 28 **(c)** 27 **(d)** 21

5. The decimal expansion of $\frac{77}{210}$ is: 1

(a) non-terminating repeating **(b)** an integer

(c) terminating **(d)** non-terminating non-repeating

6. If $\sqrt{3}$ is irrational, then $\sqrt{3} \times \sqrt{3}$ is: 1

(a) non-terminating non-repeating **(b)** irrational

(c) not defined **(d)** rational

7. The HCF of 45, 75 and 90 is: 1

(a) 16 **(b)** 17 **(c)** 14 **(d)** 15

8. The LCM of 10, 15 and 20 is: 1

(a) 61 **(b)** 300 **(c)** 60 **(d)** 30

9. The largest number that divides 58 and 98 leaving remainders 3 and 2 respectively is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

10. The least number which when divided by 4, 6 and 10 leaves remainder 2 each time is: 1

(a) 63 **(b)** 60 **(c)** 62 **(d)** 58

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the largest number which divides 245 and 1029 leaving remainders 5 and 5 respectively. 2

12. Find the smallest number which leaves remainder 5 when divided by 12, 16 and 18. 2

13. Without actual division, find how many decimal places $\frac{23}{200}$ has. 2

14. Find the largest number which divides 398 and 436 leaving remainders 7 and 11 respectively. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The HCF of two numbers is 9 and their LCM is 360. If one number is 72, find the other. 3

16. Three bells toll at intervals of 12, 15 and 18 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

17. Show that $(5 + 2\sqrt{3})(5 - 2\sqrt{3})$ is rational but $5 + 2\sqrt{3}$ is irrational. 3

OR

The HCF of two numbers is 12 and their LCM is 504. If one number is 84, find the other. 3

18. Three bells toll at intervals of 6, 10 and 14 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

OR

Show that $(7 + 3\sqrt{2})(7 - 3\sqrt{2})$ is rational but $7 + 3\sqrt{2}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two tankers contain 850 litres and 680 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 24 m by 16 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

20. Two tankers contain 612 litres and 918 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 45 m by 30 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

END OF QUESTION PAPER — Total: 40 Marks

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